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The effect of video game context and player relationships on prosocial behavior: The mediating role of reciprocal expectations

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A R T I C L E I N F O

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A B S T R A C T

Although research suggests cooperative game contexts promote prosociality, key gaps remain regarding underlying mechanisms (i.e., reciprocal expectations), moderators (i.e., player relationships), and the confounding effect of violent content in many studies. To address these issues, we conducted two experiments (N = 274) using a neutral video game to compare cooperative, competitive, and single-player contexts, measuring prosocial behavior via a give-some dilemma task. Results demonstrated that the cooperative context significantly increased prosocial behavior compared to the other two contexts, an effect partially mediated by heightened reciprocal expectations. Conversely, the competitive context did not significantly decrease prosocial behavior, and while friends were more prosocial than strangers overall, player relationship did not moderate the context's effect. These findings underscore the robust positive impact of cooperative game design and suggest that the negative effects of competition may be contingent on factors like violent content, rather than being inherent to competition itself.

1. Introduction

Video games have become a primary form of entertainment for many people, particularly college students. Over the years, the effects of Video games on individuals have been widely discussed. Research initially focused on the negative consequences, such as aggression, often explained through social-cognitive frameworks like the General Learning Model, which posits that games effectively teach whatever content and behaviors are repeatedly rehearsed [1]. However, with the rise of positive psychology in recent years, an increasing number of researchers have begun focusing on the positive effects of Video games on individuals [2]. Among these, pro-sociality, as an important component of individual socialization development, has received significant attention from researchers.

This focus on behavior within technologically mediated environments is part of a broader academic trend. For instance, researchers are identifying student behavior in smart classrooms to improve educational outcomes [3] and analyzing human behavior data to aid in the prevention of noncommunicable diseases [4]. A key insight emerging from these fields is that understanding context is crucial for the accurate recognition of individuals' affective states and behaviors [5].

Consistent with this perspective, Gentile [6] noted that Video games can affect individuals in five ways: the amount of play, game content, game context, game structure, and game manipulation. While most studies have examined the effects of game content (e.g., violent vs. prosocial), the effects of game context (e.g., competitive vs. cooperative) on individuals cannot be ignored. This paper aims to explore how different game contexts affect prosocial behavior and the mediating role of reciprocal expectations in these contexts. The subsequent sections (2 and 3) will detail the experiments designed to address the specific research questions and contributions outlined below.

1.1. Game context and prosocial behavior

According to previous studies, game context can be divided into a single-player game context and a multiplayer game context, and the multiplayer game context can be further divided into competitive and cooperative contexts [6,7,8]. Different game contexts have different

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rules and goals, and accordingly, players exhibit different behaviors in different contexts. Prosocial behavior refers to individuals' behavior that benefits others and society for whatever purpose [9], including cooperative, helping, and donation behaviors [10].

Research has explored the relationship between the game context and prosocial behavior. First, cooperative game contexts promote prosocial behavior in individuals [11,12]. Studies on violent Video games have found that players in cooperative contexts adopt more prosocial behaviors than those in competitive contexts regardless of game violence [13]. Further research found that the positive effects of cooperative contexts in promoting prosocial behavior were not only seen in in-game peers but also in subsequent out-of-game behavior with strangers. While playing a single violent Video game reduced individuals' prosocial behavior, the cooperative context increased individuals' subsequent prosocial behavior toward strangers [14]. This aligns with the design philosophy of many modern pervasive games, such as moBIO Threat, which integrates various technologies to foster real-world social interaction and collaboration among players [15].

Second, competitive contexts reduced prosocial behaviors [7]; Suni & Liu, 2019). The social context of play can be a powerful moderator of game effects. Indeed, the importance of context is so profound that even the narrative framing of in-game actions—such as committing violence for a prosocial versus an ambiguous motive—can significantly alter a game's impact on aggression and prosocial thoughts [16].

However, some studies produced different results: cooperative contexts reduced individual aggression without increasing individual cooperative behavior in Jerabeck & Ferguson's [17] study. Furthermore, no correlation was found between competitive games and individual prosocial behavior in a longitudinal study [18].

In summary, most studies use violent or prosocial Video games when exploring the influence of game contexts, and the relationship between different game contexts and individuals' prosocial behavior still requires further exploration. Considering that game content itself can influence individual pro-sociality, this study attempted to measure individual prosocial behavior in terms of both prosocial behavior toward game partners and subsequent prosocial behavior toward strangers, while controlling for game content as an influencing factor, to better examine the relationship between different game contexts and prosocial behavior.

1.2. Reciprocal expectations

Researchers have explored the relationship between the game context of Video games and players' prosocial behaviors. However, the mechanisms through which game contexts influence players' prosocial behavior still require further investigation and discussion.

Bounded generalized reciprocity (BGR) theory was originally used to explain human behavior in intergroup social situations [19], particularly in interactions between strangers, without established expectations. The theory assumes that in intergroup social interactions, individuals will behave in a way that best serves their own interests, and that individuals' expectations of reciprocal behavior from others influences their behavior in social interactions. Individuals will behave positively toward others when they believe that in-group members will behave positively, expecting the same positive feedback that maximizes their own interests [20]. However, in the case of out-group members, it is clear that the likelihood of them reciprocating positive behavior is lower, and thus individuals do not have such reciprocal expectations for out-group members [21].

As research progressed, some researchers found that BGR could also be used to explain the effects of different game contexts in Video games on individuals' prosocial behaviors. Reciprocal expectations refer to individuals' expectations of reciprocal behavior from others. According to BGR, individuals' reciprocal expectations may mediate role in the pathway of the influence of game contexts on individuals' prosocial behavior: cooperative contexts promote individuals' reciprocal expectations, which leads to positive attitudes and increased prosocial behaviour [14,11,19]. The theory states that the "team member" cue activates and inspires individuals to cooperate with in-group members [22,19], and the "team member" cue is closely related to the formation of cooperative contexts. Research has shown that cooperative contexts enable players to view game partners as team members, creating a positive norm of reciprocity that raises players' reciprocal expectations, thus leading to more positive attitudes and increased prosocial behaviour [14,11]. However, competitive game contexts can cause players to categorize competitors as out-group members who have no positive interactions with each other. In this context, players attack and obstruct each other, and may perceive that others return the same aggressive behavior, thus creating a negative norm of reciprocity and decreasing reciprocal expectations, which in turn decreases their subsequent prosocial behavior [11,12,22]. However, there is one area previous studies have not sufficiently explored: whether competitive contexts reduce prosocial behaviors through the mechanism of diminished reciprocal expectations.

Therefore, based on BGR, this study aims to investigate the role of reciprocal expectations across different Video game contexts. Specifically, it examines reciprocal expectations as a mediating variable between game contexts and prosocial behaviors, thereby elucidating the pathway through which game contexts influence player behavior.

1.3. Player relationship

In addition to "how you play" influencing individuals, "who you play with" in different game contexts may also influence prosocial behavior. In recent years, Video games have increasingly incorporated interactive and real-time multiplayer elements, enabling players to engage with others in both cooperative and competitive scenarios [23]. While many games still provide single-player modes, the growing emphasis on online interaction means that players often encounter co-players who may be friends or strangers.

This aligns with broader findings that individual player characteristics significantly shape their game experience; for example, a player's prior experience with a game genre directly influences their perceptions of its usability and appeal [24]. Similarly, the pre-existing social relationship between players is a crucial individual-level factor that may alter their behavioral responses. Since friends have established a certain degree of connection, trust, and behavioral interaction patterns beforehand, whereas strangers do not have such emotional connections [25], different types of relationships between players in games may mean individuals have different game experiences. As a result, their emotional states and behavioral responses differ.

This aligns with insights from gamification research, which suggests that systems should be designed oriented to users' motivational characteristics for optimal engagement [26]. The relationship with a co-player (friend vs. stranger) fundamentally alters a player's motivational profile in a given context. Playing games with friends elicits higher physiological arousal and more positive emotional responses than playing games with strangers [27]. Therefore, when the competition and cooperation objects of a game are replaced by friends in real life, will different game contexts have the same effect on individual prosocial behaviors as strangers? Some studies suggest that playing games with friends increases an individual's sense of commitment to goals compared with strangers in cooperative contexts. In contrast, there is no difference between the two in competitive contexts [25]. Some studies suggest no difference in prosocial behavior between friends and strangers in different game contexts, which may be because cooperative contexts promote the formation of relationships between individuals and strangers and increase their sense of trust and expectation of reciprocity between them [28]. Another study found that a competitive context can reduce the quality of friendships between individuals and their friends, but it does not affect subsequent prosocial behavior. This may be because the individuals have already established a positive relationship. Knowing that they will continue to interact positively after the game, they can still enhance their expectation of reciprocity [22]. In general, the effect of game situations on prosocial behavior may vary with different player relationships.

Prior research has indicated that both gaming contexts (cooperative vs. competitive) and player relationships (friends vs. strangers) can influence players' emotional responses and prosocial behaviors. However, how these factors jointly affect prosocial outcomes in gaming remains underexplored and warrants further investigation.

1.4. Comparison with related work and scientific contribution

While the studies reviewed in the preceding sections provide a valuable foundation, they also reveal clear gaps in the literature when examined closely. Most research has focused on comparing only two contexts at a time (e.g., cooperative vs. competitive) and has often overlooked the underlying psychological mechanisms or the social dynamics between players. Moreover, many of these influential studies have been conducted using violent Video games (e.g., [13,14], making it difficult to isolate the effects of the social context from the violent content itself. As some researchers note, utilizing a non-violent game is an important step to remove confounding explanations that could be attributed to violent content [11]. To clarify the unique contribution of the present study, the following is a systematic analysis of key related works (Table 1).

As the comparative analysis illustrates, this study makes a distinct scientific contribution by: (a) systematically compare single-player, cooperative, and competitive contexts in a unified design; (b) explicitly test the mediating role of reciprocal expectations for both the positive effects of cooperation and the negative effects of competition; and (c) investigate the moderating role of player relationship on these established pathways for prosocial behavior.

1.5. Research questions and Hypotheses

This paper adopts a two-experiment design to systematically explore the proposed variable relationships. The specific hypotheses are derived from Bounded Generalized Reciprocity (BGR) theory and the limitations of previous research. Experiment 1, as a foundational study, aims to establish the main effect of game context on prosocial behavior and test the mediating mechanism of 'reciprocal expectations' among strangers. Based on the results of Experiment 1, Experiment 2 introduces a key social moderator—player relationship (friends vs. strangers)—to examine the boundary conditions of the main effect and enhance the ecological validity of the results.

This progressive, two-step experimental approach allows for a rigorous and nuanced examination of the following core research questions and hypotheses:

Research Question 1: How does game context influence individuals' prosocial behavior?

Hypothesis 1 Compared to the single-player context, the cooperative context can increase the prosocial behavior of the individual; competitive contexts reduce prosocial behavior.

Research Question 2: Does reciprocal expectation mediate the relationship between game context and prosocial behavior?

Hypothesis 2 Compared to the single-player context, the cooperative context increases an individual's reciprocal expectations, while the competitive context decreases them.

Hypothesis 3: Reciprocity expectations mediate between game contexts and prosocial behavior.

Research Question 3: How does player relationship type moderate the impact of game context on prosocial behavior and reciprocal expectations?

Hypothesis 4 Individuals will exhibit higher reciprocal expectation and prosocial behavior when their game partners are friends than when they are strangers.

Hypothesis 5: When the game partner is a stranger, a cooperative context will increase reciprocal expectations and prosocial behavior, whereas a competitive context will decrease reciprocal expectations and prosocial behavior. When the game partner is a friend, the game context will have no significant *effect on reciprocal expectations or prosocial behavior.*

2. Experiment 1

Experiment 1 was designed as a foundational study to test the main effect of game context on prosocial behavior and the mediating role of reciprocal expectations. Specifically, this experiment aimed to provide an empirical test of Hypotheses 1, 2, and 3. The main recruitment methods for participants were campus posters and distribution of recruitment flyers. To ensure a controlled test of the core mechanism, the experiment was conducted with participants who did not previously know each other.

2.1. Materials and methods

2.1.1. Participants

Ninety-four students from a Chinese university participated in Experiment 1. Among them, 44 were boys and 50 were girls. Participants of the same sex were randomly matched in pairs. The participants

Table 1
Comparison of Key Studies on Game Context and Prosocial Behavior.

Study	Game Context Investigated	Key Findings on Prosocial Behavior	Mediators/Moderators Explored	Identified Gap / Limitation
Ewoldsen et al. [13]	Cooperative vs. Competitive in a violent game (Halo II)	Cooperative play led to more tit-for-tat strategies (a precursor to cooperation) compared to competitive play.	–	Did not test underlying psychological mechanisms.
Greitemeyer et al. [14]	Cooperative vs. Single-player in violent games (Far Cry, FlatOut)	Cooperative play increased cooperation relative to single-player mode, an effect that generalized to strangers.	Mediation: Cohesion → Trust Norms → Cooperation.	Lacked a direct comparison with a competitive context.
Velez [11]	Quality of cooperation (Helpful vs. Unhelpful teammate) in a non-violent game	Helpful teammates confirmed and unhelpful teammates disconfirmed positive reciprocal expectations, affecting prosocial behavior accordingly.	Mediation: Reciprocal Expectations.	Did not directly compare cooperative vs. competitive contexts; focused on the quality of cooperation.
Peng & Hsieh [25]	Cooperative vs. Competitive; Friends vs. Strangers in a non-violent game	Friends showed higher goal commitment in cooperative contexts; no difference in competitive contexts.	Moderation: Player Relationship.Mediation: Goal Commitment.	The dependent variable was goal commitment/ performance, not prosocial behavior.
Current Study	Cooperative vs. Competitive vs. Single-player in a neutral game	(To be tested)	Mediation: Reciprocal Expectations.Moderation: Player Relationship.	Addresses the gaps by: (1) Systematically comparing all three contexts; (2) Testing the full mediating path for prosocial behavior; (3) Examining the moderating role of player relationship on prosocial outcomes.

did not know each other and were randomly assigned to different experimental conditions. Because part of the data of three participants was lost during the experiment, effective data for 91 participants were finally obtained, comprising 31 in the single-player context, 30 in the cooperative context, and 30 in the competitive context.

The sample size was determined by a priori power analyses of G*power 3.1.9.4[29] and the maximum number of participants we could collect. Before the experiments, an a-prior power analysis (with $\alpha = 0.05$) calculated with G*power indicated that 66 participants were required to detect an expected large effect size of $f = 0.40$ [30] with 80 % power in a one-way analysis of variance (ANOVA). Finally, based on the maximum number of participants we could collect in the school, 94 participants were eventually recruited.

2.1.2. Materials

2.1.2.1. Video games. Based on the purpose of the experiment, this study selected a neutral Video game with three game contexts: single-player, cooperative, and competitive. Based on previous research, we selected "Mario Kart: Double Dash!!" (hereafter referred to as Mario Kart) as the game material. "Mario Kart" is a racing game published by Nintendo that allows players to drive go-kart cars on various tracks, obtain item boxes, and use various items to increase their speed, defend against, and block other cars. The game consists of three game contexts: single-player, cooperative, and competitive. In a single-player context, the player controls the kart and uses items alone. In the cooperative context, the two players share the game screen on a computer and operate the kart together. One player is responsible for driving the kart, and the other is responsible for using the props. The two players can switch roles at any time by pressing the buttons simultaneously. In a competitive context, two players share a computer screen. The screen is divided into two halves. Each player controls their own kart and races against the other.

2.1.2.2. Video game evaluation. A. Game Experience Questionnaire: The adapted game experience questionnaire[31] was used to ask participants to list three Video games they had played frequently in the past six months and score each game on a seven-point scale (1 = almost none, 7 = always) for the frequency of use, degree of violence, and degree of competition in the content contained in the game. Violent Video games experience = Σ [violence degree of game content × (frequency of playing games on weekdays × 5 + frequency of playing games on non-weekdays × 2) / 7] / 3, the higher the score, the more violent Video game experience the individual has; Competition Video game experience = Σ [competition degree of game content × (frequency of playing games on weekdays × 5 + frequency of playing games on non-weekdays × 2) / 7] / 3, the higher the score, the more competitive Video game experience the individual has; Cooperation Video game experience = Σ [cooperation degree of game content × (frequency of playing games on weekdays × 5 + frequency of playing games on non-weekdays × 2) / 7] / 3, the higher the score, the more cooperative Video game experience the individual has[32].

B. Game Evaluation Questionnaire: Adapted from the game evaluation questionnaire[31,7], this scale measures the level of violence, prosocial degree, frustration, difficulty, acceptance, pleasure, excitement, love, proficiency, cooperation, competition, and other content of the game, and includes 10 questions. The questionnaire uses a seven-point scale, with 1 indicating "strongly disagree" and 7 indicating "strongly agree." In addition, at the end of the questionnaire, the participants were asked to report whether they knew the game partner and to rate the degree of cooperation and competition with the game partner on a scale of 1–7, to test the effectiveness of the experimental condition manipulation.

2.1.2.3. Prosocial behavior. Based on the investigation of individual prosocial behavior in previous studies, the prosocial behavior of individuals in this study includes the prosocial behavior of game partners and the prosocial behavior of strangers outside the game. The specific measurement methods were as follows.

A. prosocial behavior of game partners: Adapted from a single trial of a two-person give-some dilemma task[22]. In the task, participants were told that they would receive 10 tickets, each of which would be worth 1 RMB. Additionally, if they divided the tickets among the other parties, they would receive 2 RMB for each ticket given to the other party, and 2 RMB for the other's ticket given to them. They could allocate as many votes as they saw fit, and the outcome of their choice would be kept completely secret. The number of tickets received by the participants was converted into money according to the rules and included in the participant's fee. The number of tickets allocated to game partners (0—10) was used to measure prosocial behavior toward the game partners.

B. the prosocial behavior of strangers outside the game: The "help in future research" contextual measure was used. After the completion of the experiment, the participants were informed that the experiment was over and given the appropriate payment for the experiment. The participants were then informed that in another laboratory, another graduate was working on a final project, and that they needed the participants to take part in the experiment. However, because it was extra work, no funding was applied for the experiment, so they could not provide a fee. The participants were given a recruitment slip to fill in with contact information and the time (in minutes) they were willing to help, with 0 indicating that they were not willing to help. The more helping time the participants volunteered, the higher their level of prosocial behavior. To improve credibility, the participants were informed that the next experiment would be arranged according to their personal wishes.

2.1.2.4. Reciprocal expectations. Based on existing research, the present study assessed participants' reciprocal expectations of game partners in a single trial of the two-person give-some dilemma task[33,14,22], i.e., participants were asked "From 0 to 10 tickets, how many do you think your game partner will give you?" The higher the number, the higher the reciprocity expectations of the individuals.

2.1.2.5. Prosocial tendencies questionnaire and Cooperative-Competitive personality tendencies questionnaire. A. The prosocial tendencies questionnaire [34] was used to assess whether participants had different prosocial traits in different experimental conditions and contained 26 items on a 5-point Likert scale from 1 (not at all like me) to 5 (very much like me). In this study, the Cronbach's $\alpha = 0.90$.

B. The cooperative-competitive personality disposition questionnaire was used to assess participants' cooperative and competitive personality traits[35]and contained 23 questions on a 9-point scale ranging from 1 (strongly disagree) to 9 (strongly agree). In this study, Cronbach's α coefficient of the cooperative personality tendency subscale was 0.87, and that of the competitive personality tendency subscale was 0.82.

2.1.3. Procedure

The participants first completed an informed consent form that outlined the background of this experiment and the problems that might be encountered in the experiment. After completing the game experience questionnaire, the cooperative-competitive personality disposition questionnaire, and the prosocial tendencies questionnaire, the participants were randomly assigned to the experimental groups and briefly informed about how the game would operate. Following a 5-minute game practice session, the participants were allowed to play the game for 15 min and fill the game evaluation questionnaire after the game. They then completed a single trial of the two-person give-some dilemma task and finally filled in the "help in future research" questionnaire.

After completing all the questions, the participants were asked whether they were aware of the true purpose of the experiment; their responses indicated that they were not. Therefore, aside from the three participants whose data were lost during the experiment, the remaining 91 participants provided valid data for subsequent analyses.

2.2. Results

2.2.1. Experimental operation inspection

A one-way ANOVA was conducted on the homogeneity of the participants in the three different game contexts and on their post-game evaluations. The results are shown in Table 2. There were no significant differences ($p > 0.05$) among the three groups in terms of game experience, prosocial tendencies, cooperative and competitive personality tendencies, or various game evaluations. The degree of competition in the competitive context was significantly higher than that in the single-player and cooperative contexts. There was no significant difference between the single-player and cooperative contexts; the degree of cooperation in the cooperative context was significantly higher than that in the single-player and cooperative contexts, and there was no significant difference between the single-player and competitive contexts.

The analysis results showed that the participants in different game contexts were homogeneous and that the game contexts in the study were grouped effectively by matching game characteristics.

2.2.2. Effects of game context on prosocial behavior, reciprocity expectations

A one-way ANOVA was conducted, with participants' prosocial behavior toward game partners, time devoted to helping strangers outside the game, and reciprocity expectations as the dependent variables (Table 3). Significant differences were found among the three game contexts in participants' prosocial behavior toward game partners ($p < 0.01$), time devoted to helping strangers outside the game ($p < 0.05$), and reciprocity expectations ($p < 0.01$).

Post-hoc comparisons by Fisher's LSD (Table 4) revealed that participants in the cooperative context showed significantly higher prosocial behavior toward game partners than in the single-player ($p < 0.001$) and competitive contexts ($p = 0.01$), but there was no significant difference between the competitive and single-player contexts. Participants in the cooperative context spent significantly more time helping strangers outside the game than those in the single-player context ($p = 0.01$), but there was no significant difference between the competitive and single-player contexts. Participants in the cooperative context had significantly higher reciprocal expectations of game partners than those in the single-player context ($p < 0.01$) or the competitive context ($p < 0.01$); however, there was no significant difference between the competitive and single-player contexts.

2.2.3. The mediating role of reciprocal expectations

First, the game context was dummy coded. The game context with the cooperative context was coded as (1,0) and the competitive context as (0,1), with the single-player context (0,0) as the reference. This was followed by a mediation model with the game context as the independent variable, the prosocial behavior of individuals toward game partners and the prosocial behavior of strangers outside the game as the dependent variables, reciprocity expectation as the mediating variable, and gender as the control variable, using Model 4 in the SPSS process 2.16.3 plug-in to perform the mediating effects test.

3. Prosocial behavior toward game partners

The mediation model for prosocial behavior toward game partners was significant. The overall total effect of game context was significant, $F_{(2,\ 87)} = 7.52$, $p < 0.01$. The direct effect was also significant, $F_{(2,\ 86)} = 4.14$, $p < 0.05$, and, importantly, the overall indirect (mediation) effect via reciprocal expectations was significant, 95 % Bootstrap CI [0.01, 0.24].

Subsequent relative mediation analysis revealed a significant indirect effect for the cooperative context (vs. single-player), 95 % Bootstrap CI [0.66, 2.45]. As illustrated in Fig. 1, participants in the cooperative context reported significantly higher reciprocal expectations than those in the single-player context (path $a_1 = 1.62$, $p < 0.01$). Higher reciprocal expectations were, in turn, associated with significantly greater prosocial behavior (path $b = 0.95$, $p < 0.001$). The indirect effect ($a_1b = 1.54$) accounted for 66.09 % of the total effect. The direct effect of the cooperative context remained significant ($c_{1'} = 0.79$, $p = 0.03$), indicating a partial mediation. In contrast, for the competitive context, the indirect effect was not significant, 95 % Bootstrap CI [-1.23, 0.86].

4. Prosocial behavior toward strangers

For prosocial behavior toward strangers, the overall total effect of game context was significant, $F_{(2,\ 87)} = 4.27$, $p < 0.05$. However, the overall indirect effect through reciprocal expectations was not significant, 95 % Bootstrap CI [-0.09, 0.49]. This indicates that reciprocal expectations did not mediate the relationship between game context and prosocial behavior directed at strangers.

In conclusion, reciprocal expectations acted as a significant mediator only for prosocial behavior directed toward game partners. The cooperative context boosted this behavior both directly and indirectly by

Table 2
Matching results of game features in three game contexts ($M \pm SD$).

	Single-player	Cooperative	Competitive	F	p	η^2
Violent video game experience	8.28 ± 6.07	10.38 ± 8.71	7.79 ± 5.71	1.18	0.31	0.03
Competitive video game experience	13.35 ± 6.95	14.27 ± 8.08	12.07 ± 6.02	0.73	0.48	0.02
Prosocial tendencies	3.64 ± 0.53	3.65 ± 0.45	3.63 ± 0.42	0.02	0.98	0.00
Cooperative personality tendencies	6.43 ± 0.96	6.17 ± 0.99	6.27 ± 1.28	0.43	0.65	0.01
Competitive personality tendencies	4.68 ± 1.30	4.59 ± 1.36	4.64 ± 1.30	0.04	0.96	0.00
Degree of violence	1.84 ± 1.16	1.63 ± 0.93	2.00 ± 1.31	0.77	0.46	0.02
Degree of prosocial	1.68 ± 0.79	1.90 ± 1.13	1.73 ± 1.05	0.41	0.67	0.01
Frustration	3.94 ± 1.77	3.23 ± 1.61	3.17 ± 1.86	1.82	0.17	0.04
Difficulty	4.65 ± 1.58	4.33 ± 1.47	3.90 ± 1.54	1.82	0.17	0.04
Pleasure	4.65 ± 1.23	4.93 ± 1.46	4.77 ± 1.36	0.35	0.71	0.01
Acceptance	5.13 ± 1.52	5.47 ± 1.28	5.57 ± 1.38	0.82	0.44	0.02
Proficiency	3.71 ± 1.32	3.57 ± 1.59	3.77 ± 1.79	0.13	0.88	0.00
Excitement	4.97 ± 1.20	4.77 ± 1.46	4.90 ± 1.37	0.18	0.84	0.00
Love	4.48 ± 1.29	4.53 ± 1.55	4.47 ± 1.28	0.02	0.98	0.00
Degree of competition	2.23 ± 2.00	1.47 ± 0.94	5.00 ± 1.46 ^{a***b***}	44.24***	0.00	0.50
Degree of cooperation	1.71 ± 1.58	5.57 ± 1.46^{b***c***}	1.93 ± 1.08	73.44***	0.00	0.63

Note: [a] "competitive context" compared with "single-player context"; [b] "competitive context" compared with "cooperative context"; [c] "single-player context" compared with "cooperative context".

Table 3
Comparison of differences in prosocial behavior of participants in three game contexts (M ± SD).

	Single player	Cooperative	Competitive	*F*	*p*	η^2
Prosocial behavior of game partners (Tickets)	3.65 ± 1.91	5.93 ± 2.55	4.30 ± 2.74	7.20**	0	0.14
Prosocial behavior of strangers (Time)	14.52 ± 14.28	25.83 ± 14.86	22.00 ± 16.43	4.38*	0.02	0.09
Reciprocal expectations	3.71 ± 1.77	5.30 ± 2.17	3.50 ± 2.47	6.29**	0	0.13

Note:$^*p < 0.05$, $^{**}p < 0.01$.

Table 4
Post-hoc comparisons (LSD) of prosocial behaviors and reciprocal expectations.

Dependent Variable	(I) Game Context	(J)Game Context	Mean Difference (I-J)	*SE*	*p*
Prosocial behavior of game partners (Tickets)	Cooperative	Single player	2.29***	0.62	0.00
	Cooperative	Competitive	1.63*	0.63	0.01
	Competitive	Single player	0.66	0.62	0.29
Prosocial behavior of strangers (Time)	Cooperative	Single player	11.32**	3.89	0.01
	Cooperative	Competitive	3.83	3.93	0.33
	Competitive	Single player	7.48	3.89	0.06
Reciprocal expectations	Cooperative	Single player	1.59**	0.55	0.01
	Cooperative	Competitive	1.80**	0.56	0.00
	Competitive	Single player	−0.21	0.55	0.71

Note: Mean Difference is calculated as (I) context minus (J) context; $^*p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

enhancing reciprocal expectations. This mediating pathway was not observed for behavior toward strangers or in the competitive context, as detailed above and visualized in Fig. 1 for the partner-directed model.

4.1. Discussion

Experiment 1 found that the cooperative Video game context increased individuals' prosocial behavior toward game partners and strangers outside the game, whereas the competitive context did not decrease individuals' prosocial behavior. The cooperative context in the game increased individuals' reciprocal expectations for unfamiliar game partners, but the competitive context did not decrease individuals' reciprocal expectations. Reciprocal expectations mediated the relationship between game context and prosocial behaviors toward game partners. Hypothesis 3 (Reciprocity expectations mediate between game contexts and prosocial behavior) was supported, and Hypotheses 1 (Compared to the single-player context, the cooperative context can increase the prosocial behavior of the individual; competitive contexts reduce prosocial behavior) and Hypotheses 2 (Compared to the single-player context, the cooperative context increases an individual's reciprocal expectations, while the competitive context decreases them) were partially supported.

4.1.1. The influence of game context on prosocial behavior

The results showed that the prosocial behaviors of the participants in the cooperative context were significantly higher than of those in the single-player context. However, the competitive context group did not differ significantly from the single-player context group in these two aspects, indicating that the cooperative context promoted the prosocial behaviors of individuals. From the results of existing studies, regardless of whether the game contains violent or prosocial content, the cooperative context can promote individuals' prosocial behavior. This positive effect is not only reflected in the face of game partners in the game, but also carries over to prosocial behavior toward strangers outside the game[36,13]. The results of the present study validate the positive effects of cooperative contexts in neutral Video games.

However, the study did not find a negative effect of competitive context on prosocial behavior toward game partners or out-of-game strangers, similar to the findings of Lobel et al. [18]. This study suggests that the reason for this result, in addition to the possibility that the negative effects of competition may be moderated by the interactive fun of a mildly competitive game context with friendly competitors., may also be related to an individual's level of moral cognition where their moral code and value judgments influence their behavior. It has been shown that an individual's moral judgment significantly predicts their prosocial behavior, and the higher the moral judgment ability, the more prosocial behavior an individual exhibits[37]. Therefore, owing to their own moral constraints, individuals' prosocial behavior is not significantly reduced, even in competitive contexts. In addition, frequency and length of play may influence the effects of competitive contexts (Langlois et al., 2023). One study found that competitive play was associated with a decrease in prosocial behavior for individuals who played Video games frequently[38]. In the present study, individuals only played for 15 min, and the short duration of play may not be sufficient to have a significant negative effect on individuals' cognition, emotion, and behavior in competitive contexts.

4.1.2. The effect of game context on reciprocal expectations and the mediating role of reciprocal expectations

BGR suggests that in cooperative contexts, positive interactions between players cause players to categorize partners as team members and create reciprocal expectations, and individuals then behave positively toward team members in the expectation of receiving the same positive feedback. In contrast, competitive contexts cause players to view competitors as out-group members, and out-group members do not exhibit positive behaviors in their favor, and thus individuals do not develop reciprocal expectations[22]. The results of this study show that individuals' reciprocal expectations do differ significantly across the three game contexts, with individuals in the cooperative context generating significantly higher reciprocal expectations for their game partners than in the single-player and competitive contexts. In contrast, there is no

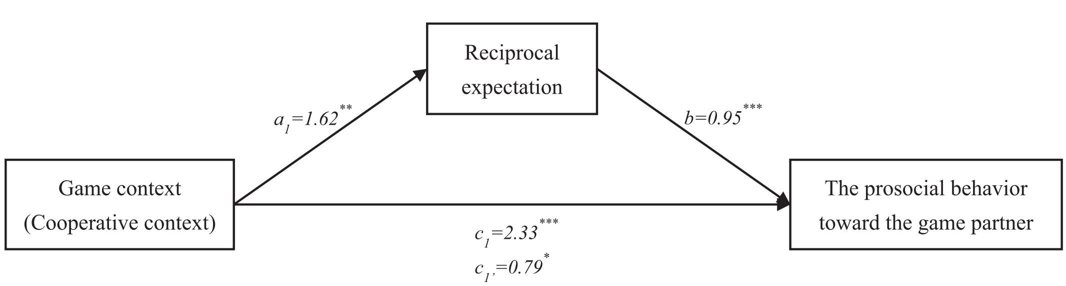

Fig. 1. Path diagram of the mediating effect of reciprocal expectations between game situations and pro-social behavior of game partners.

significant difference between the single-player and competitive contexts, suggesting that cooperation does raise individuals' reciprocal expectations for their game partners, but competition does not lower individuals' reciprocal expectations. This finding is consistent with those of previous studies[21], and may be related to the direct face-to-face involvement of players in games. According to the intergroup contact hypothesis, direct intergroup contact under specific conditions can facilitate the development of coordinated intergroup relationships [39]. In this study, the players and their competitors in the competition context were both racing on the same computer. Although the players did not communicate with each other, this face-to-face participation may have established a better sense of connection between the players compared to those who played the game alone. In the competition situation, the players were not only competitors but also partners in the game together, which may buffer the negative effects of competition on individuals.

It was also found that reciprocal expectations partially mediated the effect between the cooperation context and prosocial behavior toward game partners. However, the mediating effect of reciprocal expectations between the competition context and prosocial behavior toward game partners was not significant, nor was there a mediating effect between the game context and individuals' prosocial behavior toward strangers. This result suggests that cooperative contexts in Video games not only directly affect individuals' prosocial behavior toward game partners but also promote individuals' reciprocal expectations toward game partners, leading to an increase in prosocial behavior, which is consistent with previous findings[33,11,21].

These findings support BGR and emphasize the important role of reciprocity expectations in individual behavior. Competitive contexts did not reduce individuals' reciprocity expectations, and therefore, did not lead to a decrease in prosocial behavior. Because information about the relationship between competitive contexts and individuals' reciprocal expectations and prosocial behavior is sparse and controversial, more research is needed to investigate this issue. Furthermore, although some studies have shown that individuals' reciprocal expectations formed during play carry over even to their behavior toward strangers [11,21], no effect of reciprocal expectations on individuals' prosocial behavior toward strangers was found in the present study. This may be because the game duration was only 15 min in this study. Furthermore, the facilitative effect of the game context and reciprocal expectation was more reflected in individuals' prosocial behavior toward their direct contact with their game partners. However, the extent of this influence was not enough to change individuals' perceptions, attitudes, and behaviors toward strangers outside the game. Therefore, the positive effect of reciprocal expectation on game partners did not carry over to individuals' prosocial behavior toward strangers outside the game.

5. Experiment 2

Building on the findings of the initial study, Experiment 2 was designed to examine the role of player relationship, investigating how the type of relationship between players (friends vs. strangers) moderates the effects of game context. Specifically, this experiment aimed to test the interaction effects proposed in Hypotheses 4 and 5. The main recruitment methods were campus posters and recruitment flyers, which invited two types of volunteers: (a) pairs of same-sex friends who signed up together to participate in the 'friends' condition, and (b) individuals who signed up alone and were subsequently paired with another unknown, same-sex participant for the 'strangers' condition.

5.1. Materials and methods

5.1.1. Participants

In Experiment 2, participants were asked to invite their real-life same-sex friends to participate in the experiment. One hundred and eighty participants from the same university participated in Experiment 2. Among them, 82 were boys, and 98 were girls. Participants were randomly placed in same sex pairs, with some participants paired with friends and others paired with strangers and then randomly assigned to three contexts: single-player (30 in the friends group and 32 in the strangers group), competitive (28 in the friend's group and 30 in the stranger's group), and cooperative (30 in the friend's group and 30 in the stranger's group). Owing to missing experimental data of one participant in the single-player friend group during the experiment, valid data for 179 participants were finally obtained.

The sample size was determined by a priori power analyses of G*power 3.1.9.4[29]and the maximum number of participants we could collect. Before the experiments, an a-prior power analysis (with $\alpha = 0.05$) calculated with G*power indicated that 64 participants were required to detect an expected large effect size of $f = 0.40$[30]with 80 % power in a 3 × 2 analysis of variance (ANOVA). Finally, based on the maximum number of participants we could collect in the school, 180 participants were eventually recruited.

5.1.2. Materials

The experimental materials were the same as those used in Experiment 1. The difference being that individuals' prosocial behavior toward strangers outside the game was measured using a donation task[40]as follows: participants were shown a donation initiative (fictitious content). Participants could voluntarily donate a portion of the reward, and the higher the donation amount, the higher their level of prosocial behavior.

5.1.3. Procedure

The study procedure was the same as that in Experiment 1. The difference is that the "help in future research" questionnaire filled out after the game was changed to a donation task.

5.2. Results

5.2.1. Experimental operation inspection

A one-way ANOVA was conducted on the post-game evaluations of the participants in the three game contexts, and the results are shown in Table 5. There were no significant differences ($p > 0.05$) among the three groups of participants in all game evaluations, except for frustration ($p < 0.01$). Therefore, frustration was included as a control variable in the subsequent analyses. As for the degree of competition, the degree of competition in the competition context was significantly higher than that in the single-player and cooperation contexts, and there was no significant difference between the single-player and cooperation contexts. The degree of cooperation in the cooperation context was significantly higher than that in the single-player and cooperation contexts, and there was no significant difference between the single-player and competition contexts. Taken together, the matching results of the game characteristics indicate that the three groups of game contexts in the experimental study were grouped effectively, and the game characteristics were matched.

Statistical analysis of participants' scores on game experience, prosocial tendencies, and cooperative-competitive personality tendencies was conducted using a 3-play context (single-player/cooperative/competitive) × 2-player relationship (friend/stranger) two-factor between-subjects ANOVA. The results showed no significant differences ($p > 0.05$) in participants' game experience, prosocial tendencies, and cooperative-competitive personality tendencies under different experimental conditions, and additional effects of game experience, prosocial tendencies, and cooperative-competitive personality tendencies on the experimental results could be excluded.

5.2.2. Effects of game context and player relationship on reciprocal expectations and individual prosocial behavior

Frustration was used as a control variable and a 3-play context (single-player/cooperative/competitive) × 2-player relationship

Table 5
Matching results of game features in three game contexts ($M \pm SD$).

	Single player	Cooperative	Competitive	F	p	η^2
Degree of violence	1.64 ± 1.02	1.58 ± 0.83	1.62 ± 1.11	0.05	0.95	0.00
Degree of prosocial	1.82 ± 1.30	2.07 ± 1.22	1.74 ± 0.98	1.24	0.29	0.01
Frustration	4.11 ± 1.86 ^{a**c*}	3.42 ± 1.79	3.00 ± 1.60	6.15**	0.003	0.07
Difficulty	4.46 ± 1.61	4.48 ± 1.44	4.00 ± 1.39	1.98	0.14	0.02
Pleasure	4.70 ± 1.45	5.10 ± 1.36	5.17 ± 1.31	2.00	0.14	0.02
Acceptance	5.38 ± 1.34	5.58 ± 1.25	5.78 ± 1.31	1.39	0.25	0.02
Proficiency	3.44 ± 1.54	3.45 ± 1.63	3.52 ± 1.58	0.04	0.96	0.00
Excitement	4.62 ± 1.40	5.00 ± 1.50	4.91 ± 1.34	1.18	0.31	0.01
Love	4.36 ± 1.29	4.93 ± 1.38	4.74 ± 1.32	2.91	0.06	0.03
Degree of competition	2.44 ± 1.83	1.88 ± 1.53	4.91 ± 1.64 ^{a***b***}	54.69***	0.00	0.38
Degree of cooperation	1.77 ± 1.38	4.88 ± 1.81^{b***c***}	1.69 ± 1.08	93.46***	0.00	0.52

Note: [a] "competitive context" compared with "single-person situation"; [b] "competitive context" compared with "cooperative context"; [c] "single-player context" compared with "cooperative context".

(friend/stranger) two-factor between-subjects ANOVA was used to statistically analyze participants' reciprocity expectations and prosocial behavior (see Table 6).

The results indicate that in terms of reciprocal expectations, there is a marginal significant main effect of the game context ($p = 0.05$). In the cooperative context, participants' reciprocal expectations ($M = 6.39$, $SE = 0.29$) were significantly higher than in the competitive context ($M = 5.39$, $SE = 0.30$). There is a significant main effect of player relationship ($p < 0.001$), where participants in the friend condition had significantly higher reciprocal expectations ($M = 7.33$, $SE = 0.24$) compared to the stranger condition ($M = 4.33$, $SE = 0.24$). The interaction between game context and player relationship was not significant ($p > 0.05$).

In terms of prosocial behavior of game partners, there is a significant main effect of the game context ($p < 0.01$). In the cooperative context, participants' prosocial behavior of game partners ($M = 7.02$, $SE = 0.29$) is significantly higher than in the single-player context ($M = 5.95$, $SE = 0.29$) and the competitive context ($M = 5.71$, $SE = 0.30$). However, there is no significant difference between the competitive and single-player contexts. There is a significant main effect of player relationship ($p < 0.001$), where participants in the friend condition showed significantly higher prosocial behavior of game partners ($M = 7.61$, $SE = 0.24$) compared to the stranger condition ($M = 4.84$, $SE = 0.24$). The interaction between game context and player relationship was not significant ($p > 0.05$).

In terms of prosocial behavior of strangers, the main effect of game context, the main effect of player relationship, and the interaction effect between game context and player relationship were not significant ($p > 0.05$).

5.3. Discussion

The results showed that after controlling for frustration, individuals' reciprocal expectations and prosocial behaviors toward their game partners were significantly higher in the cooperative context than in the single-player context. While there was no significant difference between the competitive and single-player contexts, the reciprocal expectations and prosocial behaviors were significantly higher in the friend's group than in the stranger's group. This result supports Hypothesis 4, whereas Hypothesis 5 is only partially supported. The results also reaffirm the findings of Experiment 1 that cooperative contexts promote individuals' reciprocal expectations and prosocial behaviors toward game partners, whereas competitive contexts do not significantly affect reciprocal expectations and prosocial behaviors. Because individuals in the stranger group did not know each other before the game, while players in the friend group did, it was not surprising that they continued to get along closely with each other in real life after the game. Therefore, the reciprocal expectations and prosocial behavior of the friend group were higher than those of the stranger group were. However, this result differs from the existing research in that there is no significant difference between the cooperative behaviors of friends' and strangers' in different game contexts [28]. More research is needed to discuss this, as very few studies have explored the differences in individuals' behavior toward friends and strangers during play and the factors that influence them.

In addition, Experiment 2 showed that individuals in different game contexts and player relationships did not differ significantly in their prosocial behaviors toward strangers outside of play, which is inconsistent with previous research [40]. The study speculates that the applicability of the measure of prosocial behavior outside the game may need to be considered. In the present study, individuals' subsequent out-of-game prosocial behavior was measured using a donation task. Although this measure has been used in existing studies, the willingness and amount of donation by the participants in the actual measure of the present study may be influenced by a number of factors outside the experiment, such as the participants' financial status and their trust in the donor organization unit. For most individuals, donation behavior is based on the premise that they can survive on their own, and that their economic status is an important factor in determining the occurrence of donation behavior[41]. Since all participants were college students,

Table 6
Game contexts and player relationship on reciprocal expectations and prosocial behavior.

Factor	Dependent Variable	SS	df	MS	F	p	η^2
Frustration	Reciprocal expectation	2.23	1	2.23	0.44	0.51	0.00
	Prosocial behavior of game partners	13.11	1	13.11	2.59	0.11	0.02
	Prosocial behavior of strangers	5.86	1	5.86	0.14	0.71	0.00
Game context	Reciprocal expectation	31.11	2	15.56	3.08*	0.049	0.04
	Prosocial behavior of game partners	57.57	2	28.79	5.70**	0.00	0.06
	Prosocial behavior of strangers	79.20	2	39.60	0.95	0.39	0.01
Player relationship	Reciprocal expectation	389.91	1	389.91	77.15***	0.00	0.31
	Prosocial behavior of game partners	334.02	1	334.02	66.09***	0.00	0.28
	Prosocial behavior of strangers	116.61	1	116.61	2.81	0.10	0.02
Context × Relationship	Reciprocal expectation	4.33	2	2.17	0.429	0.65	0.01
	Prosocial behavior of game partners	12.42	2	6.21	1.23	0.30	0.01
	Prosocial behavior of strangers	132.59	2	66.30	1.60	0.21	0.02

Note:*$p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

their primary financial support generally came from their families. Participation in the study itself served as a means to earn extra income. For those with limited financial means or strained economic conditions, the participation fee may have represented a meaningful source of income. Consequently, donation willingness and amount may not have accurately or directly reflected the participants' underlying prosocial tendencies.

6. General discussion

Experiment 1 explored the effects of the game context in Video games on individuals' prosocial behavior. The results showed that the cooperative context did promote prosocial behavior, which was consistent with the hypothesis; however, the competitive context did not have a significant effect on prosocial behavior, which was not consistent with the expected hypothesis. In addition, the study also verified the important mediating role of reciprocal expectation between game context and prosocial behavior. This partially verified Hypothesis 3, in that cooperative contexts promote not only prosocial behavior toward game partners but also reciprocal expectation toward game partners, which further increases prosocial behavior. This result is another verification of the BGR. However, the competitive context did not reduce individuals' reciprocal expectations, and the mediating role of reciprocal expectations between the competitive context and individuals' prosocial behaviors did not hold.

Based on Experiment 1, Experiment 2 further explored whether there were differences in the effects of the game context on individuals' prosocial behaviors based on the type of relationship between players. The results showed that the effect of game context on individuals' prosocial behavior was essentially the same regardless of whether the game partner was a friend or stranger, in that cooperative contexts could enable individuals' reciprocal expectations and prosocial behavior toward game partners, while competitive contexts did not significantly affect these outcomes. This result validates the results of Experiment 1. However, regardless of context, individuals' reciprocal expectations and prosocial behaviors toward friends were significantly higher than those of strangers were. This result means Hypotheses 4 and 5 are partially supported.

6.1. Re-evaluating the effects of competition

The results of this study prompted us to ask: In an era when online Video games have become the main form of entertainment for many people, are the negative effects of competitive contexts in Video games being amplified? First, it has been debated in related studies whether violent content or competitive contexts have a negative impact on individuals. Some researchers have argued that the negative effects of violent content in games on individuals are limited, the effect is exaggerated, and that the competitive element of the game, not the violent one, brings about negative effects[42,43,44]. However, in the present study, competitive context did not have a negative effect on individuals either. Since most previous studies chose violent Video games or prosocial Video games in the selection of experimental materials, the results obtained may be due to the joint effect of game content and game context. In contrast, the present study used neutral Video games, which excluded the interference of game content to a certain extent, thus leading to differences in the study results. The speculation that there may be an interaction between game content and game context–that is, the competitive context has a significant negative effect on individuals only when combined with violent content–needs to be substantiated by further research.

Second, the influence of competitive games on individuals may also be influenced by other aspects, such as the motivation of individuals to play games or the characteristics of competitive objects. Surveys have shown that socialization has become one of the main motivations for adolescents to play games [45,46]; therefore, both cooperation and competition are forms of individual participation in games and interaction with others[47]. In this context, the discomfort associated with competition is weakened, enhancing the enjoyment of the game and increasing opportunities for individuals to interact with others. At the same time, friendly competitors[48], and individuals' own higher moral judgment[37], all to a certain extent, can turn a seemingly aggressive competition into a fun interactive contest.

Finally, although a large body of research has focused on the negative effects of competition, several studies have identified its positive effects. For example, competitive games are important markers of healthy child development, forcing children to interact benignly while following common rules [49]. In a longitudinal study, competitive games were associated with a decrease in children's problem behaviors and an improvement in peer relationships [18]. Several studies have also shown that adolescents who play competitive and cooperative online games show more stable emotional states and good peer relationships [50], and that competitive games can reduce stress effectively [51].

Thus, the results of our experiments suggest that cooperation factors have positive effects on individuals, but the effects of competition factors are not all negative. Therefore, when exploring the negative effects of competition, we should also consider various aspects to avoid magnifying them.

6.2. Limitations of the research

While the present study provides valuable insights, several limitations should be acknowledged when interpreting the results. First, the participants were all university students from a single cultural background, which may limit the generalizability of the findings to other age groups or cultures. Second, prosocial behavior was measured using a laboratory-based dilemma task and a helping scenario; while these are established methods, their direct correspondence to real-world, long-term prosocial behavior requires further investigation. Third, this study utilized a single neutral Video game to control for content effects. Future research could explore whether these findings hold across different genres of neutral games (e.g., puzzle, racing, strategy). Finally, this research focused on the immediate, short-term effects of game play. The long-term consequences of sustained exposure to cooperative or competitive game contexts remain an open question.

6.3. Summary of key findings

To distill the primary contributions of this work, the key findings and their implications are summarized in Table 7.

7. Conclusion

This study investigated the effects of different game contexts on individuals' prosocial behavior in Video games, examined the intrinsic mechanism of reciprocity expectation, and explored whether these effects differed across players' relationships. Our results confirmed that cooperative contexts significantly increase individuals' prosocial behaviors toward game partners and strangers, an effect partially mediated by heightened reciprocal expectations. In contrast, and contrary to our initial hypothesis, competitive contexts in a neutral Video game did not significantly reduce individuals' prosocial behaviors or their reciprocal expectations. Furthermore, while individuals consistently demonstrated higher reciprocal expectations and prosocial behaviors toward friends than strangers across all game contexts, the positive effect of the cooperative context itself did not significantly differ between relationship types.

7.1. Future work

The findings and limitations of this study open several avenues for future research. First, a crucial next step is to directly test the interaction

Table 7
Summary of Key Findings and Lessons Learned.

Key Finding / Lesson Learned	Implication / Explanation
1. Cooperation reliably promotes prosociality.	A cooperative game context significantly increases both reciprocal expectations and subsequent prosocial behavior, an effect that holds even for strangers. This confirms the robustness of cooperation as a positive social mechanism.
2. Competition is not inherently negative.	In a neutral game, a competitive context did not significantly decrease prosocial behavior or reciprocal expectations. This challenges the common assumption that competition always leads to negative social outcomes.
3. Game content may be a critical moderator.	The neutral effect of competition in our study, contrasted with negative effects found in studies using violent games, suggests that the negative impact of competition may only emerge or be amplified when paired with violent content.
4. Pre-existing friendships create a higher baseline for prosociality.	Individuals are consistently more prosocial toward friends than strangers across all contexts. However, the effect of the game context itself (i.e., cooperation's positive boost) was not significantly different for friends versus strangers.
5. Reciprocal expectation is a key mechanism.	The study confirms that the positive effect of cooperative contexts on prosocial behavior is mediated by an increase in players' expectations that their partners will reciprocate positively.

between game context and game content. An experiment comparing cooperative and competitive play in both neutral and violent games would clarify whether the negative effects of competition are indeed contingent on violent content. Second, longitudinal studies are needed to understand the long-term effects of habitual cooperative or competitive gameplay on individuals' social skills and relationships. Finally, future research could explore other potential moderators, such as players' personality traits (e.g., trait competitiveness, agreeableness) and different game mechanics, to build a more comprehensive model of how Video games shape social behavior.

In summary, the results of this study suggest that cooperative contexts positively affect individuals' prosocial behaviors. However, the effects of competitive contexts on individuals are complex and should not be measured only by the polarization of "good" and "bad." In the era of the rapid development of online Video games, whether individuals compete with strangers or friends does not mean that they will be negatively affected by competition; therefore, the impact of competition factors on individual psychology and behavior should be viewed dialectically.

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CRediT authorship contribution statement

Mingchen Wei: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jie Lin:** Methodology, Investigation, Formal analysis, Data curation. **Xu Wang:** Writing – review & editing. **Shuai Chen:** Writing – review & editing. **Yanling Liu:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethical Approval

All study participants provided informed consent. The study has been approved by the appropriate ethics review board (H23129) and has been performed in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki and its later amendments.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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